Tr. Julius W. M **0787093081**

Candidate's Name: ……SAMPLE SCORE……..

Signature: ……..

STREAM:							

525/1
CHEMISTRY
Paper 1
Nov./Dec.
2025
$2\frac{1}{2}$ hours

END OF YEAR ASSESSMENT 2025

Uganda Advanced Certificate of Education

S.5 CHEMISTRY

Paper 1

(***Set II***)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper has **two** *sections* **A** *and* **B.**

Section **A** *has two compulsory items while* **B** *has two parts* **Part I** *and* **Part II**

Each of part I and part II has two items, Answer only **one** *from each.*

Answers to Section **A** ***must*** *be written in the spaces provided and Section* **B** ***must*** *be written in the answer booklet(s) provided*

Answer four in all.

Where necessary use,

Molar gas volume at s.t.p = 22.4dm^3

FOR EXAMINER'S USE ONLY				
ITEM	**CODE**		**SCORE**	
1	F_1		04	
	C_1		04	
	C_2		06	
	C_3		03	
	D_1		03	
2	E_c		06	
	C_4		02	
	C_5		04	
	E_1		05	
	D_2		03	
3/4	G_r	V_p	05	08
	T_r	R	08	09
	E_v	D_3	04	03
	P_r		03	
5/6	F_2		06	
	Mc		02	
	S		11	
	D_3		03	
TOTAL			80	

Key

C_1, C_2, C_3, C_4, C_5: *Calculations*
D_1, D_2, D_3: *Danger*
E_c: *Energy Cycle*
E_V: *Evaluation*
G_r: *Graphing*
T_r: *Trend*
P_r: *Pridiction*
F: *Formula*
Mc: *Mechanism*
S: *Synthesis*
V_p: *VSEPR theory*
R: *Reason*

SECTION A

Answer **all** *questions in this section*

ITEM 1

During a rapid response investigation near Kitezi landfill, the Water Quality Surveillance Unit discovered heavy metal contamination in a borehole supplying a large number of households.

A fragment of metallic lead from a nearby illegal battery recycling site was analysed using a mass spectrometer. The detector registered four isotopic signals with currents of *0.16 mA (mass 204), 2.72 mA (mass 206), 2.50 mA (mass 207)* and *5.92 mA (mass 208)*. The investigation team was briefed that these readings can help determine an accurate relative atomic mass (RAM) of lead.

Further laboratory tests on the contaminated water revealed that the dissolved poisonous salt responsible for the lead toxicity contained *62.6%* lead, *8.45%* nitrogen, and the rest oxygen molar mass of the salt was reported as *331.22* g/mol.

The concentration of the dissolved salt in the borehole water was measured as 0.90 mol dm^{-3}, and the borehole supplies approximately 1200 litres of water per day. Chemists concluded that this salt can be removed by treating the water with sodium sulphate solution, which reacts to produce insoluble lead(II) sulphate.

In extreme contamination cases, *0.45 moles* of nitrogen dioxide gas was released during side reactions at 28°C and 98 kPa. To protect public health, the Ministry has assigned your team to analyse the data and provide a complete stoichiometric and methodological evaluation of the situation.

Task:

As a learner of Chemistry, help the company to know;

(a) The relative atomic mass of lead. **(RAM)**.

$$\textbf{Total current} = 0.16 + 2.72 + 2.50 + 5.92 = 11.3mA$$ ✓CA

$$\textbf{Percentage abundance of lead} - 204 = \left(\frac{0.16}{11.3} \times 100\right) = 1.42$$

$$\textbf{Percentage abundance of lead} - 206 = \left(\frac{2.72}{11.3} \times 100\right) = 24.07$$

$$\textbf{Percentage abundance of lead} - 207 = \left(\frac{5.92}{11.3} \times 100\right) = 22.12$$

✓CA

$$\textbf{Percentage abundance of lead} - 208 = \left(\frac{2.72}{11.3} \times 100\right) = 52.40$$

$$\textbf{R.A.M} = \sum \textbf{Relative isotopic mass} \times \textbf{proportion of isotope}$$

$$\textbf{R.A.M} = \left(204 \times \frac{1.42}{100}\right) + \left(206 \times \frac{24.07}{100}\right) + \left(207 \times \frac{22.12}{100}\right) + \left(208 \times \frac{52.4}{100}\right)$$ ✓CA

$$\textbf{R.A.M} = 207.26$$ ✓CA

$4CA = C_1 = 04$ *scores*

(b) To determine the molecular formula of the poisonous salt in water.

$Lead = 62.6\%, Nitrogen = 8.45\%, Oxygen = 100 - (62.6 + 8.45) = 28.95\%$ CA

Element	Pb	N	O
%composition	62.6	8.45	28.95
Moles	$\frac{62.6}{207.26} = 0.302$	$\frac{8.45}{14} = 0.604$	$\frac{28.95}{16} = 1.809$ CA
Simplest ratio	$\frac{0.302}{0.302} = 1$	$\frac{0.604}{0.302} = 2$	$\frac{1.809}{0.302} = 6$

$Empirical\ formula\ of\ salt\ is\ PbN_2O_6$ F

$(PbN_2O_6)n = 303.3$

$(207.26 \times 1n) + (14 \times 2n) + (16 \times 6n) = 331.22 \quad \therefore n = 1$

$Molecular\ formula\ of\ the\ salt\ is\ Pb(NO_3)_2$ F Name: $Lead(ii)nitrate$ NA

$2CA + 2F + NA = F_1 = 04\ Scores$

(c) To calculate the mass of lead(II)sulphate formed if all the dissolved salt was removed

$0.9moll^{-1}means;\ 1litre\ of\ borehole\ water\ contain\ 0.9moles\ of\ Pb(NO_3)_2$ CA

$1200litres\ of\ borehole\ water\ contain\ \left(\frac{0.9\times1200}{1}\right)$ CA

$= 1080moles\ of\ Pb(NO_3)_2$

$From\ equation: Pb(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow PbSO_4(s) + 2NaNO_3(aq)$ CA

$1mole\ of\ Pb(NO_3)_2\ produces\ 1mole\ of\ PbSO_4$

$1080moles\ of\ Pb(NO_3)_2\ produce\ \left(\frac{1}{1}\times1080\right)moles\ of\ PbSO_4. = 1080moles$ CA

$Molar\ mas\ of\ PbSO_4\ is\ (207 \times 1) + (32 \times 1) + (16 \times 4) = 303g$ CA

$1mole\ of\ PbSO_4\ weigh\ 303g$

$\therefore 1080moles\ of\ PbSO_4\ weigh\ \left(\frac{303\times1080}{1}\right) = 327240g$ CA $or\ 3.27 \times 10^5 kg\ of\ PbSO_4$ CA

$7CA = C_2 = 06\ scores$

(d) Know the volume of gas produced using the gas law.

$From\ gas\ law: PV = nRT; P = 98 \times 10^3 pa, T = (28 + 273) = 301k\ and\ R = 8.314$ CA

Note: *We use*;

1. $R = 0.0821\ \text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K}), when, P\ is\ in\ atm\ and\ V\ is\ in\ litre/dm^3$
2. $R = 8.314\ J/(molK), when, P\ is\ in\ Pa\ and\ Volume\ is\ in\ m^3$

$\therefore V = \frac{nRT}{P} = \left(\frac{0.45\times8.314\times301}{98\times10^3}\right)$ CA

$= 0.0115m^3$ CA

$3CA = C_3 = 03\ scores$

(e) Environmental impacts of the gaseous or aqueous system and their mitigations.

Nitrogen dioxide contributes to ***global warming*** *when it accumulates in the atmosphere, since it traps heat and creating the greenhouse effect, leading to rising temperatures hence* ***drought****. This can be mitigated by using catalytic converters to reduce it into less harmful substances such as nitric acid.*

Nitrogen dioxide causes ***respiratory problems*** *and* ***poor air quality.*** *When inhaled, it irritates the lungs and reduces oxygen uptake, increasing cases of* ***asthma*** *and* ***bronchitis.*** *Using personal protective gears like gas masks during scientific investigations.*

$di + de + dm = D_1 = 03scores$

Item 2

A recent technical brief from Namanve Metals & Ceramics Corporation (NMCC), a rapidly expanding Ugandan manufacturer of refractory linings and water-treatment chemicals, revealed that the company is designing a *high-temperature resistant* coating for industrial furnaces used in steel rolling mills in Jinja and Tororo.

The project requires evaluating aluminium oxide and sodium fluoride, in order to make best choice of the most stable substance during formation to apply in *ceramics, abrasives,* and *furnace linings*.

At the same time, to select a substance in which temperature control depends on dissolution in water to use as a *cooling and stabilising agent* in chemical preparation units. The research division has sourced pure aluminium from scrap recycling operations in Namanve and oxygen gas from Uganda Oxygen Ltd. Sodium and fluorine also have been sourced from Lake Katwe in Kasese.

Thermochemical data obtained from an industrial materials report show the following values

$Al(s) \rightarrow Al(g)$: +330 kJ mol^{-1}	$Na(s) \rightarrow Na(g)$: $+109\ kJ\ mol^{-1}$
$Al(g) \rightarrow Al^{+}(g) + e^{-}$: +578 kJ mol^{-1}	$Na(g) \rightarrow Na^{+}(g) + e^{-}$: $+494\ kJ\ mol^{-1}$
$Al(s) \rightarrow Al^{2+}(g) + 2\,e^{-}$: +1820 kJ mol^{-1}	$F_2(g) \rightarrow 2F(g)$: $+158\ kJ\ mol^{-1}$
$Al(s) \rightarrow Al^{3+}(g) + 3\,e^{-}$: +2750 kJ mol^{-1}	$F(g) + e^{-} \rightarrow F^{-}(g)$: $-328\ kJ\ mol^{-1}$
$2Al^{3+}(g) + 3O^{2-}(g) \rightarrow Al_2O_3(s)$:−15346.5kJ mol^{-1}	$Na(s) + \frac{1}{2}F_2(g) \rightarrow NaF(s)$: $-569\ kJ\ mol^{-1}$
$\frac{1}{2}O_2(g) \rightarrow O(g)$: +249 kJ mol^{-1}	$Na^{+}(g) + (aq) \rightarrow Na^{+}(aq)$: $-406\ kJmol^{-1}$
$O(g) \rightarrow O^{-}(g)$: −142 kJ mol^{-1}	$F^{-}(g) + (aq) \rightarrow F^{-}(aq)$: $-505\ kJmol^{-1}$
$O(g) \rightarrow O^{2-}(g)$: +798 kJ mol^{-1}	

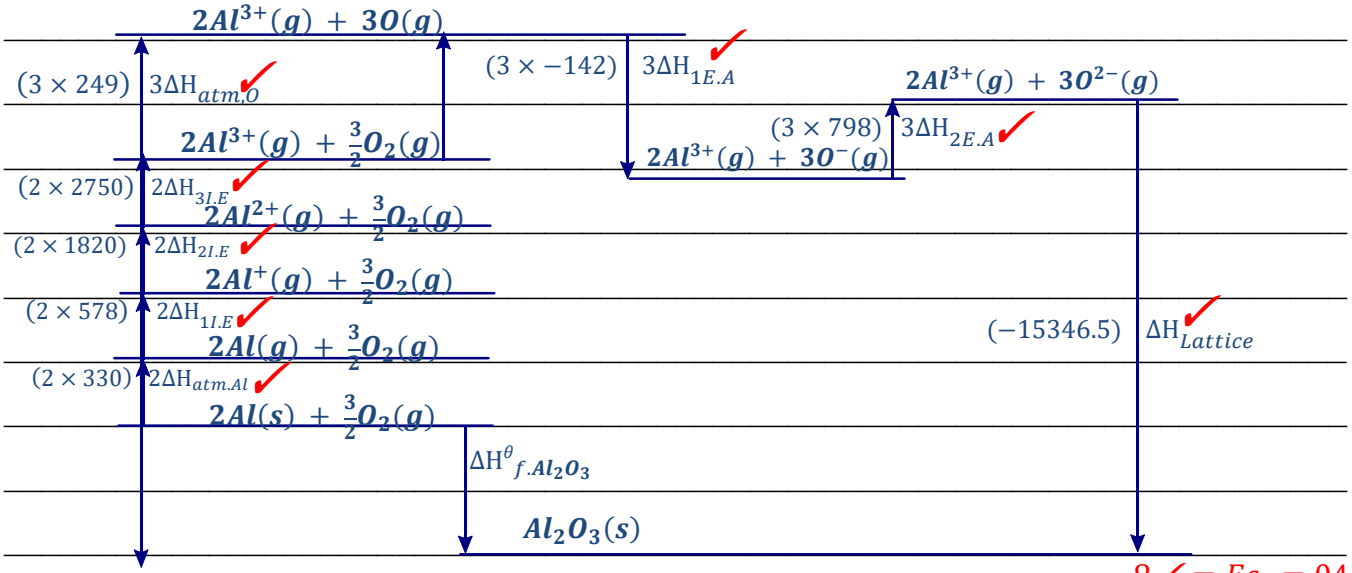
Preliminary tests by NMCC's laboratory team indicated that dissolution of NaF produces a measurable temperature change. To ensure that the chosen materials are safe, energy-efficient, and suitable for industrial use, the technical manager has invited you.

Task

As a student of chemistry, help the manager;

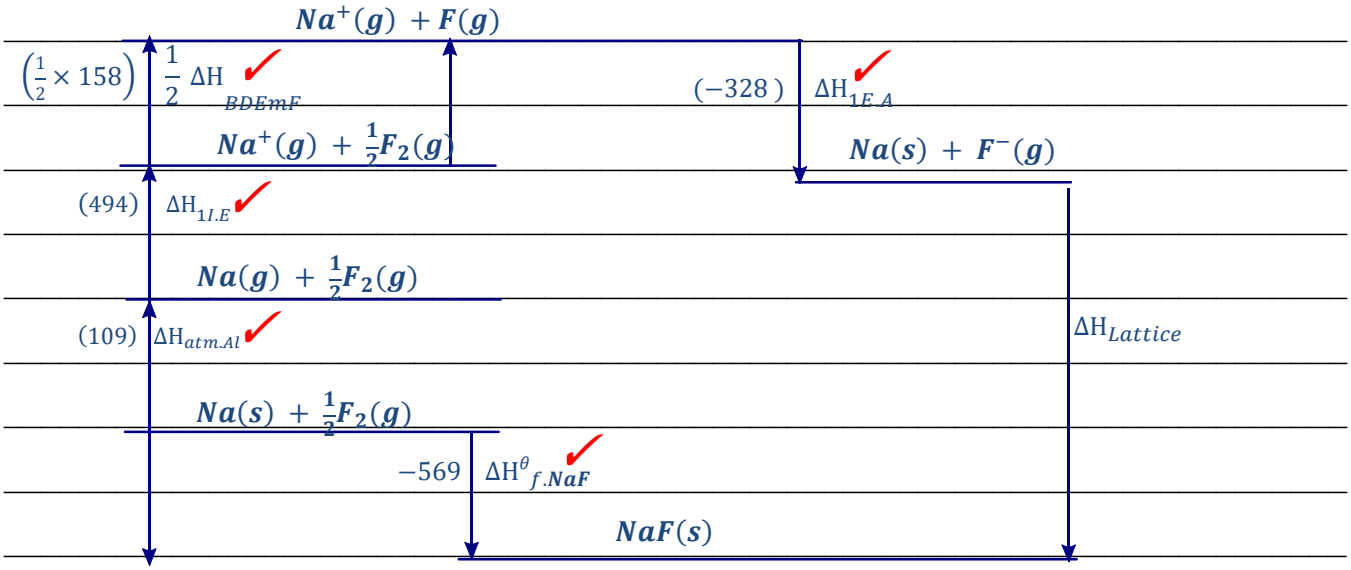
(a) Analyse the data to construct an energy level diagram for the changes in the:

(ii) Formation of aluminium oxide.



$8\checkmark = Ec_1 = 04scores$

(i) Formation of sodium fluoride.



$5 \checkmark = Ec_2 = 02 scores$

$Ec_1 + Ec_2 = E_c = 06\ scores$

(b) Use the energy cycles constructed to calculate;

(i). Enthalpy of formation of aluminium oxide.

From Hess' law;

$$\Delta H^{\theta}_{f.Al_2O_3} = 2\Delta H_{atm.Al} + 2\Delta H_{1I.E} + 2\Delta H_{2I.E} + 2\Delta H_{3I.E} + 3\Delta H_{atm,O} + 3\Delta H_{1.E.A} + 3\Delta H_{2E.A} + \Delta H_{Lattice}$$

$$\Delta H^{\theta}_{f.Al_2O_3} = (2 \times 330) + (2 \times 578) + (2 \times 1820) + (2 \times 2750) + (3 \times 249) + (3 \times -142) + (3 \times 798) + (-15346.5) \quad CA$$

$$\therefore \Delta H^{\theta}_{f.Al_2O_3} = -1675.5 kjmol^{-1} \quad CA$$

$2CA = C_4 = 02\ Scores$

(ii). Lattice energy of sodium fluoride and its enthalpy of solution.

From Hess' law;

$$\Delta H^{\theta}_{f.NaF} = \Delta H_{atm.Al} + \Delta H_{1I.E} + \frac{1}{2}\Delta H_{BDE,F} + \Delta H_{1.E.A} + \Delta H_{Lattice}$$

$$\Delta H_{Lattice} = -569 - \left(109 + 494 + \frac{1}{2} \times 158 - 348\right) = -923 kjmol^{-1} \quad CA \quad CA$$

$$\Delta H_{Solution.NaF} = \Delta H_{Lattice} + \Delta H_{hydration.NaF}$$

$$\Delta H_{Solution.NaF} = \Delta H_{Lattice.NaF} + \left(\Delta H_{hydration.Na^+} + \Delta H_{hydration.F^-}\right)$$

$$\Delta H_{Solution.NaF} = +923 + (-406 + -505) \quad CA$$

$$\therefore \Delta H_{Solution.NaF} = +12 kjmol^{-1} \quad CA$$

$4CA = C_5 = 04\ Scores$

(c) (i) Evaluate the industrial usefulness of the two systems to advise the manager which substance is most suitable for furnace lining and cooling applications, giving reasons.

Aluminium oxide *is the best choice for ceramics, abrasives and furnace linings due to* ***a very large exothermic enthalpy of formation*** *of* $-1675.5 kjmol^{-1}$*, thus extremely* ***thermal dynamically stable, heat-resistant*** *and* ***chemically inert****.*

Whereas:

Sodium fluoride *can be choosen for cooling applications due to it* ***endothermic enthalpy solution*** *of* $+12 kjmol^{-1}$ *, meaning it* ***absorbs a heat from surrounding equipment when dissolving in water****. However, its cooling effect is small because of low heat absorption.*

(ii). State any other important industrial use of aluminium oxide.

- *Used as an* ***electrical Insulator*** *in electrical and electronic devices; due to good electrical resistance and heat stability.*
- *Production of* ***Aluminium*** *Metal during the electrolytic processes due to its high purity .*
- *Used as a* ***catalyst in petroleum refining*** *and chemical manufacture due to its high surface area and chemical stability.*

$2Ev + 2Er + 1U = E_{v1} = 05\ Scores$

(d) Suggest the possible impacts and mitigations of the substances to the environment.

- *Aluminium Oxide can* ***contaminate soil*** *if waste powder poorly disposed. Since fine particles are insoluble they settle in soil,* ***altering pH*** *reduce plant yields. Mitigated by recycling waste powders and using dust filter systems.*
- *Sodium Fluoride leads to* ***water pollution*** *when in high amounts. Since fluoride ions are* ***mobile and toxic to plants, aquatic organisms, and wildlife, and can corrode infrastructure****. Mitigated by proper wastewater treatment before releasing it to surrounding water streams.*

$1di + 1de + 1dm = D_2 = 03\ Scores$

SECTION B

Part I

Attempt **One** *item in this section*

Item 3

A manufacturing company in Uganda, Victoria Industrial Chemicals Ltd, produces industrial grade materials for construction, ceramics, and high-temperature processes. The company has been advised that substances suitable for furnace linings, kiln refractory materials, and heat-resistant construction components must have high thermal stability and bonding character.

The engineering department is evaluating two types of available compounds from Period 3 elements: oxides and chlorides, to select the most suitable materials for industrial applications. The baseline requirement is that a suitable substance must remain solid above 1400 °C, must have strong ionic bonding, and be mechanically stable under operational conditions. The data provided for Period 3 oxides and chlorides are shown below:

period	**Period 3 chlorides and oxides**							
Chlorides Oxides	NaCl Na_2O	$MgCl_2$ MgO	$AlCl_3$ Al_2O_3	$SiCl_4$ SiO_2	PCl_3 P_4O_6	PCl_5 P_4O_{10}	S SO_3	Cl_2 Cl_2O_7
Melting pt of chlorides/°C	808	714	192	-68	-92	160	-76	-80
Melting pt of Oxides/°C	1275	2827	2007	1607	24	560	30	−91

The engineering team requires a comparative analysis of oxides and chlorides to determine which compounds are most suitable to serve the purpose.

They seek guidance on which properties make a substance "good", and how to evaluate *graphs* in melting points of the available compounds to explain *trends* in order to guide *selection* of suitable compounds for the intended industrial purpose, and to *predict* the likely melting point and industrial suitability of an oxide of element X between Al_2O_3 and SiO_2.

Task

As a chemistry student, make a write up you will use to help the company.

Item 4

Kilembe Kilembe Chemical Works Ltd, Kasese, is studying how bonding and molecular structure affect the properties and industrial uses of compounds in fertilizers, water purification, and plastics and ceramics manufacture. The compounds include water, ammonia, phosphorus trichloride, phosphorus pentachloride, carbon tetrachloride, tin(II) chloride, tin(IV) chloride, and nitrate ions. It was reported that some, like carbon tetrachloride and phosphorus trichloride, are liquids and insoluble in water, while water and ammonia are highly soluble with strong intermolecular attractions. Tin(II) chloride conducts electricity when molten, whereas tin(IV) chloride does not.

The manager seeks information on *molecular shape*, *bond type*, and *bond angle*, and how these explain bonding type *account for differences* in melting point, solubility, and electrical conductivity. *Compare* boiling points and solubility of water and ammonia to those of phosphorus trichloride, phosphorus pentachloride and carbon tetrachloride. *Compare* electrical conductivity and melting points of tin(II) and tin(IV) chloride. And evaluate environmental impacts of nitrate ions in fertilizers. You have been contacted for assistance.

Task;

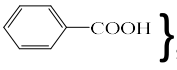
As a chemistry student, make a write up you will use to help the company.

Part II

Attempt **One** *item in this section*

ITEM 5

Nile Chemical Industries Ltd, a chemical manufacturing firm located in Jinja, produces a range of organic-based materials used in plastics, fuels, solvents, and disinfectants. In one of its experiments, 20.0 cm^3 of a gaseous hydrocarbon W was exploded with 150 cm^3 of excess oxygen. After complete combustion, the residual gas measured 110 cm^3, and when concentrated potassium hydroxide solution was added, the volume further decreased to 30 cm^3

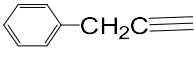
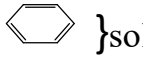
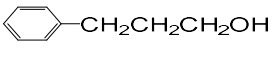
The company noted that ozonolysis of W produced only one compound and that W readily decolourised bromine water. These findings prompted the research team to seek a deeper understanding of W and its possible industrial applications in the synthesis of propan-1-ol, benzoic acid, {  }, 1,2-dibromopentane and 1-chlorobutane.

The company seeks a comprehensive evaluation: to determine the *formulae* of W, draw and name all possible *isomers*, *identify* W with reason(s), and illustrate the mechanism for the reaction between W and bromine water. Additionally, confirm the identity of the gas absorbed by potassium hydroxide solution, *predict* the functional groups of resulting organic compounds, describe *synthetic* pathways to related products, and assess the *environmental* impacts of producing above compounds in industry. You have been contacted for assistance.

Task:

As a student of Chemistry, prepare a presentation you will use upon invitation.

ITEM 6

Albertine Oil Fractionation Ltd, an industrial chemical company in the Albertine region, produces fuels, solvents, and organic feedstocks for plastics and chemical synthesis. During a research study, an organic compound P was isolated from a distillation fraction. Complete combustion of a sample of P yielded 8.8 g carbondioxide and 1.8 g water, while 0.100 g of P, vapourised at 273°C and 734 mmHg, occupied 4.46×10^{-2} dm^3. Ozonolysis of P followed by hydrolysis produced a mixed Q of two different compounds, and P was observed to react readily with bromine and concentrated sulphuric acid. The company is exploring industrial applications of these compounds for synthesis of prop-2-yn-1-ylbenzene {  } a component in manufacture of new drugs, benzene {  }solvent, and {  }chemical feedstocks.

The manager seeks to know the formulae of P, all possible structural *isomers*, *identity* of P with reasons, deduce the structure and functional groups of mixture Q, illustrate *mechanisms* for reactions of P with *bromine* and *concentrated sulphuric acid*, , outline *synthetic* pathways from Albertine oil fractions P to synthesis industrial compounds stated, and evaluate the environmental impacts of producing compounds in industry and their mitigations. You have been invited to help the manager.

Task:

As a student of Chemistry, prepare a presentation you will use to help the company.

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Te 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Ti 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

			139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 68	173 Yb 70	175 Lu 71
			227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

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(to be fastened together with other answers to paper)

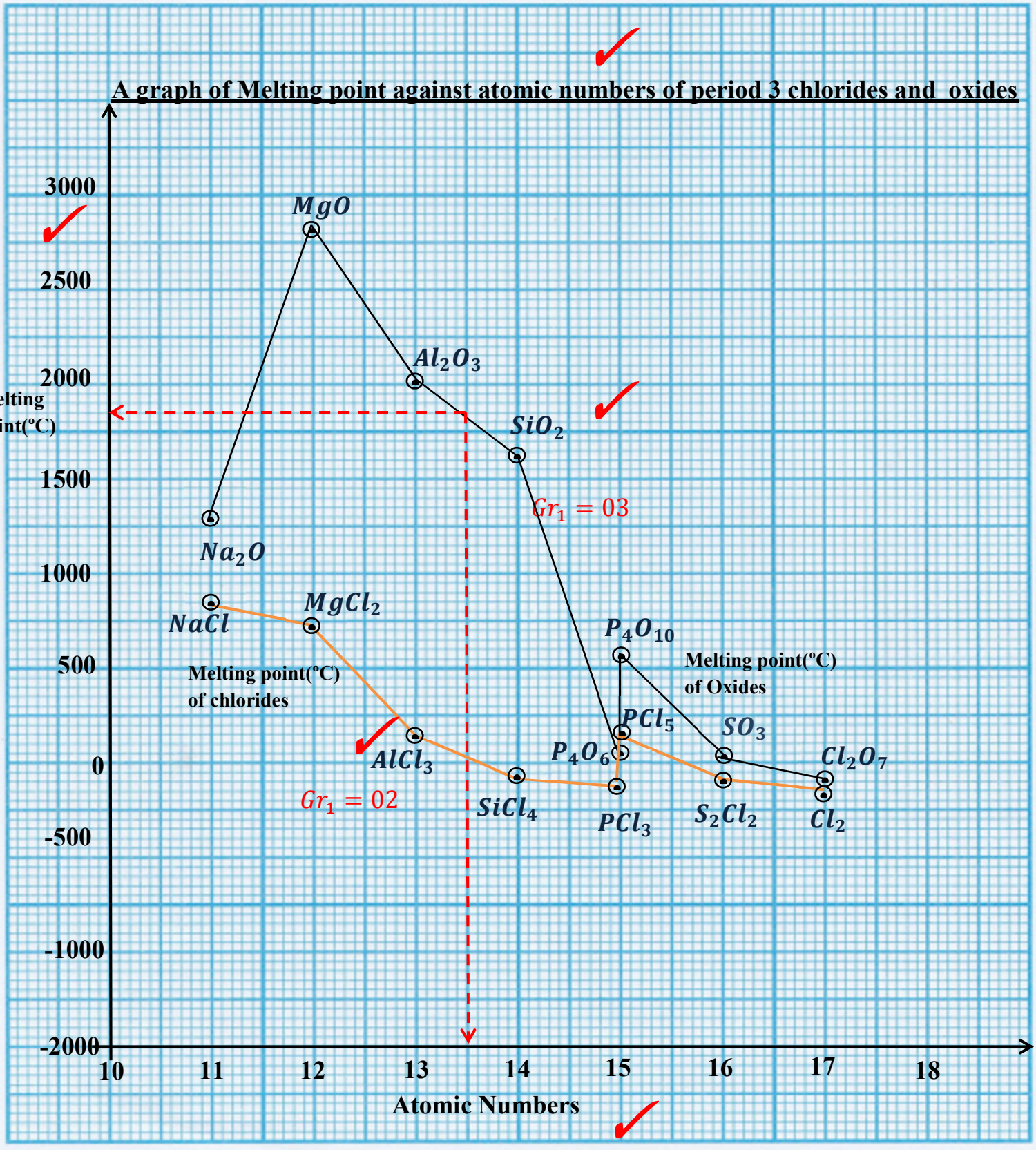
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$Gr_1 + Gr_2 = Gr = 05score$

3. **Trend in Melting points across period 3 oxides** R_2 Tr_1

Generally, melting point ***increases*** *from sodium oxide to magnesium oxide and then* ***reduces*** *to Chlorine (VII) oxide.*

Reason; R_1

Sodium oxide, magnesium oxide and aluminium oxide have ***giant ionic structures*** *whose ions are held by* ***strong ionic bonds****. Melting point increases* R_1 *from sodium oxide to magnesium oxide because magnesium oxide has a* ***higher product of charges*** *than sodium oxide.* R_1 *Melting point* ***decreases*** *from magnesium oxide to aluminium oxide because the* ***aluminium ion has a small ionic radius, very high charge density*** *and* ***high polarising power****, making aluminium oxide* ***less ionic****. Silicon(IV) oxide has a* ***giant covalent structure*** R_1 *in which each silicon atom is bonded to four oxygen atoms making* ***very many strong covalent bonds*** R_1 *that require a* ***high amount of energy to break****. Phosphorus pentoxide, sulphur trioxide and Chlorine(VII) oxide have* ***simple molecular structures*** *whose molecules are held by* ***weak van der waal's forces of attraction whose strength decreases with decrease in polarity of the molecules*** R_1 *as a result of increase in electronegativities of the atoms bonded to oxygen.*

Irregularities;

MgO has ***higher melting point*** *than* Na_2O *becau it has* ***stronger ionic bonding*** R_1 *due to* ***smaller, doubly charged ions****, giving it a much* ***higher lattice energy*** *than* Na_2O*, whose ions carry only* ***single positive charges*** *and have larger interionic distances.*

P_4O_{10} *has higher melting point than* P_4O_6*, beucause it has a* ***larger molecular size*** R_1 *and* ***stronger van der Waals forces*** *that* ***require more energy to be overcome*** *than those in the smaller* P_4O_6 *molecule.*

Trend in Melting points across period 3 chlorides;

Melting point of the chlorides generally ***decrease*** Tr_2 *from sodium chloride to dichloride.*

Reason;

Sodium chloride and magnesium chloride have ***giant ionic structures*** R_2 *held by* ***strong ionic bonds*** R_2 *which require a* ***high amount of energy to break****. The decrease in melting point from sodium chloride to magnesium chloride is because magnesium ion has a* ***smaller ionic radius****,* ***higher charge density*** *and* ***higher polarising power*** R_2 *than sodium ion making magnesium chloride* ***less ionic*** *than sodium chloride. Aluminium chloride has a* ***lower melting point*** R_2 *than magnesium chloride because among the cations, the aluminium ion has the* ***smallest ionic radius****, a* ***very high charge density*** *and* ***high polarising power****. Aluminium chloride is therefore predominantly* ***covalent****. Silicon tetrachloride is* ***covalent*** *and its molecules are held by* ***weak Van der Waals' forces of attraction that require a low amount of energy to break*** R_2 *hence a low melting point. Phosphorus trichloride has a* ***simple molecular structure*** R_2 *thus a low melting point. Disulphur dichloride and dichloride form simple molecular structures held by weak Van der Waals' forces of attraction.*

Irregularities;

*Phosphorus pentachloride has an **abnormally high** melting point because at ordinary temperatures it consists of PCl_4^+ + and PCl_6^- ions hence exhibiting ionic character.* R_2 R_2

$Tr_1 + Tr_2 + 4Tr_1 + 4Tr_2 = T_r = 08scores$

Evaluation and choice;

*Both period 3 **metal** oxides and chlorides exhibit **ionic** characters, due to their **very high melting points**, which indicate **strong electrostatic forces** between metal cations and chloride anions.* E_S

However,

*The Period-3 **oxides** are far better substances for furnace linings, kiln refractories and heat-resistant construction than the chlorides because oxides such as **MgO (m.p. 2827 °C)**, **Al_2O_3 (2007 °C)** and **SiO_2 (1607 °C) remain solid** well above the 1400 °C baseline and are **thermal dynamically stable**.* E_d E_d

*They have high melting points due to **very large lattice energies** and **strong network covalent bonding.***

Whereas;

*The metal **chlorides** (NaCl 808 °C), ($MgCl_2$ 714 °C),($AlCl_3$ 192 °C), and the molecular chlorides of non-metals have **low melting points** and are **volatile** because they are **simple covalent molecules**; therefore they **fail the >1400 °C requirement** and unsuitable for high-temperature structural use.* E_d E_d

$1E_S + 3E_d = E_v = 04scores$

Prediction of melting point, nature and industrial suitability of chloride X;

*The oxide of element **X**, lying between Al_2O_3 and SiO_2 will likely have a melting point between 1607 °C and 2007°C, approximately 1850°C.* Pr

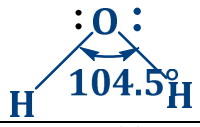
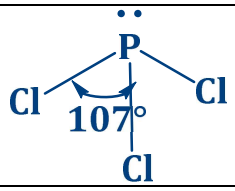
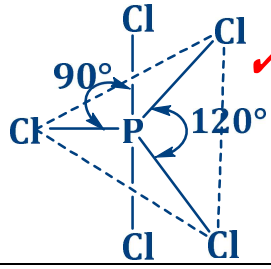
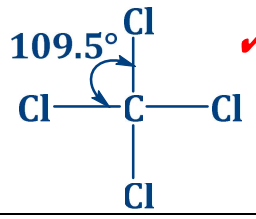
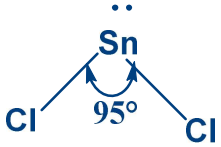
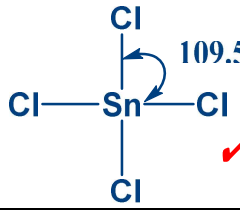
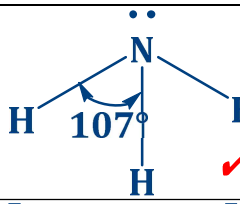
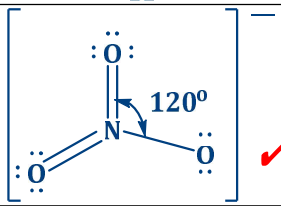
*Its bonding is expected to bean oxide with mixed **covalent and ionic** bonding rather than a purely ionic or simple molecular structure.* Pr

Industrial suitability:

*such an oxide would be **suitable for high-temperature applications** (furnace linings, kiln refractories, and heat-resistant ceramics) because it meets the >1400 °C requirement and offers good thermal stability.* Pr

$3Pr = P_r = 03scores$

4. Molecular shape, bond type, and bond angle;

Name of substance	Molecular structure	Name of molecular shape	Bond type	Bond angle
Water	 ✓Mi	*bent shape/ V-shape* ✓Ni	*Polar covalent* ✓Bi	104.5° ✓Ai
Phosphorus trichloride	 ✓Mi	*trigonal pyramidal* ✓Ni	*Polar covalent* ✓Bi	107° ✓Ai
Phosphorus pentachloride	 ✓Mi	*trigonal bipyramidal* ✓Ni	*Covalent* ✓Bi	90° and 120°. ✓Ai
Carbon tetrachloride	 ✓Mi	*tetrahedral* ✓Ni	*Non-polar covalent* ✓Bi	109.5°. ✓Ai
Tin(II) chloride	 ✓Mi	Bent shape/ V-shape ✓Ni	*Ionic with covalent character* ✓Bi	95° ✓Ai
Tin(IV) chloride	 ✓Mi	*tetrahedral* ✓Ni	*Pure covalent* ✓Bi	109.5°. ✓Ai
Ammonia	 ✓Mi	*trigonal pyramidal* ✓Ni	*Polar covalent* ✓Bi	107° ✓Ai
Nitrate ion	 ✓Mi	*trigonal planar* ✓Ni	*Covalent with resonance delocalization* ✓Bi	120° ✓Ai

$9Mi + 9Ni + 9Bi + 9Ai = V_p = 08 scorers$

Account for differences in melting point, solubility, and electrical conductivity;

*The **differences in solubility, melting point, and conductivity** among these compounds are due to the type of bonding and molecular structure.*

Ionic compounds *($SnCl_2$, nitrate salts) have **high melting points** and **conduct electricity when molten or in solution**, while **covalent molecular compounds** (CCl_4, PCl_3, $SnCl_4$) have **low melting points**, are **non-conductors**, and are **insoluble** in water.*

Hydrogen bonding *in water and ammonia leads to **strong intermolecular attractions**, giving them higher boiling points than similar covalent molecules like PCl_3 and PCl_5.*

$3R_3 = R_3 = 03scores$

Comparison of boiling points and solubility of water and ammonia to those of phosphorus trichloride, phosphorus pentachloride and carbon tetrachloride;

*Water is a **bent polar molecule** with an angle of **104.5°**, and ammonia is **trigonal pyramidal** with an angle of **107°**; both have **polar covalent bonds** and form **hydrogen bonds** between molecules. Hydrogen bonding gives them **high boiling points** and makes them **highly soluble in water**.*

*In contrast, **PCl_3** (trigonal pyramidal, 107°) and **PCl_5** (trigonal bipyramidal, 90° & 120°) are **covalent** and **cannot form hydrogen bonds**. They have **low melting points**, are **liquid or volatile solids**, and **do not dissolve** in water because they hydrolyse, producing **HCl** and **phosphoric acids**.*

$3R_4 = R_4 = 03scores$

Comparison of electrical conductivity and melting points of tin(II) and tin(IV) chloride;

*Tin(II) chloride ($SnCl_2$) is **ionic with some covalent character**, forming a **bent structure** (≈95°) and a **solid** at room temperature. It **conducts electricity when molten** because it contains **mobile ions (Sn^{2+} and Cl^-)**.*

*Tin(IV) chloride ($SnCl_4$), on the other hand, is **tetrahedral (109.5°)** and **purely covalent**, existing as a **molecular liquid** that **does not conduct electricity**. Its weak molecular forces give it a **lower melting point** than $SnCl_2$.*

$3R_5 = R_5 = 03scores$

$R_3 + R_4 + R_5 = R = 09scores$

Evaluation of the environmental impacts of using nitrate ions in fertilizers;

*In fertilizer use, **nitrate ions (NO_3^-)** improve plant growth.*

*However when overused, cause **environmental pollution** since Nitrates are **highly soluble** and can **leach into groundwater**, causing **eutrophication** and **oxygen depletion** in water bodies, leading to fish deaths and **blue-baby syndrome** in humans. Mitigated by controlled, fertilizer application, and use of organic manure.*

$1di + 1de + 1dm = D_3 = 03\ Scores$

5. **Formulae of W ;**

$C_xH_y(g) + \left(x + \frac{y}{4}\right) O_2(g) \longrightarrow xCO_2(g) + \frac{y}{2}H_2O(g)$

Volume of hydrocarbon $= 20cm^3$

Volume of carbon diode produced $= 110 - 30 = \mathbf{80cm^3}$

Volume of oxygen used up in reaction = original volume − volume unreacted

$= 150 - 30 = \mathbf{120cm^3}$ ✓CA

Volume of hydrocarbon $\times x =$ *volume of carbondioxide*

$\Rightarrow 20x = 80$

$\therefore x = {}^{80}/_{20} = \mathbf{4}$

Also volume of hydrocarbon $\times \left(x + \frac{y}{4}\right) =$ *volume of oxygen used up*

$\Rightarrow 20\left(4 + \frac{y}{4}\right) = 120$

$\therefore \ y = (6 - 4) \times 4 = 8$ ✓CA

$\therefore$ ***The formula of W is*** C_4H_8 ✓CA

Possible isomers, identity of W with reason(s)

$CH_3CH_2CH = CH_2$

$CH_3CH = CHCH_3$ ✓CA

Since ozonolysis of W produced only one compound, it indicates that it has a symmetrical carbon-carbon double bond.

$CH_3CH = CHCH_3 \xrightarrow[2.\,H_2O,\,Zn,\,CH_3COOH]{1.\,O_3,\,CCl_4,\,< 20^0C} 2CH_3CHO$ ***, One compound produced is ethanal***

Therefore, W is $CH3CH = CHCH3$ ***but-2-ene*** ✓CA

Comfirmatory Test for Carbon dioxide; ✓I

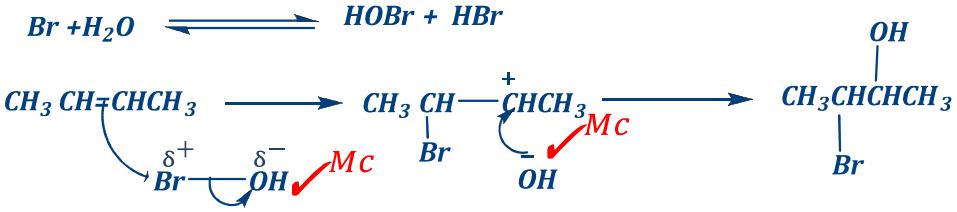
If the gas is CO_2*, KOH will absorb it so the* ***limewater remains clear*** *(no milky cloud), because* CO_2 *was removed by the KOH beforehand. But the gas is not absorbed by KOH, the limewater will still* ***go milky****.* ✓I

$5CA + 2I = \text{F} = 06$ *Scores*

Mechanism for reaction between W and bromine water;

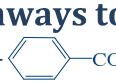
$CH_3CH = CHCH_3 \xrightarrow{Br_2/H_2O} CH_3CHBrCH(OH)CH_3 + HBr$

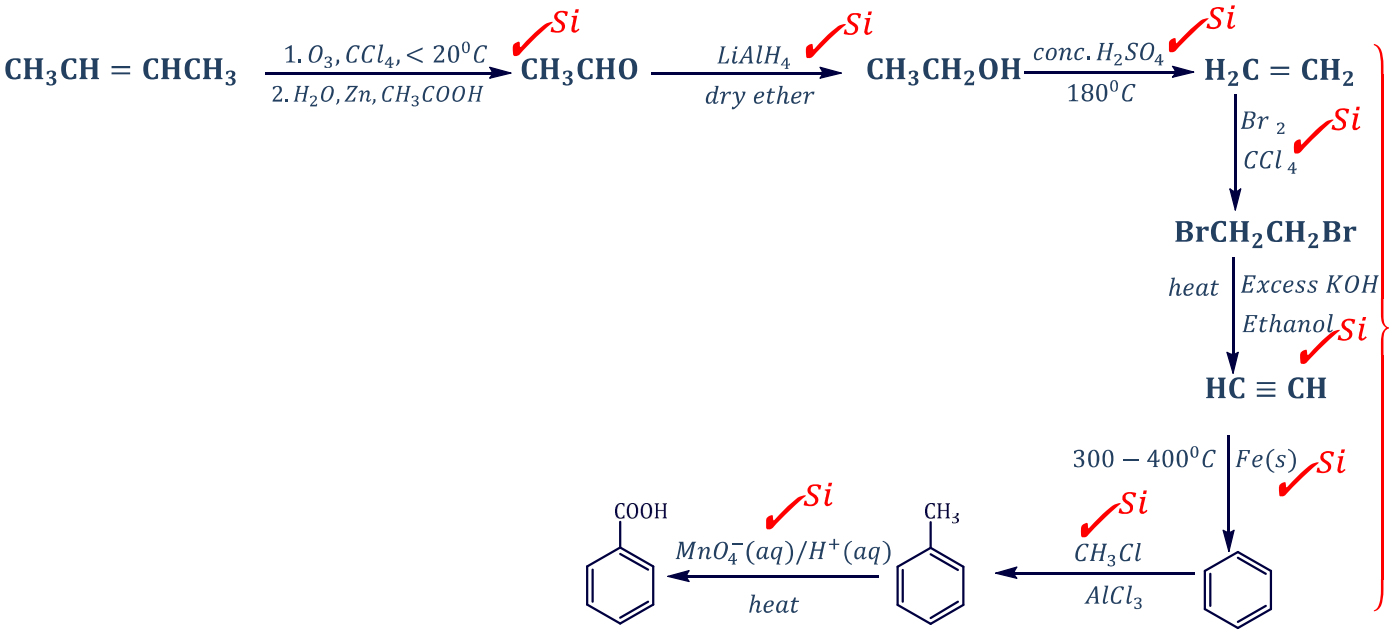
Mechanism



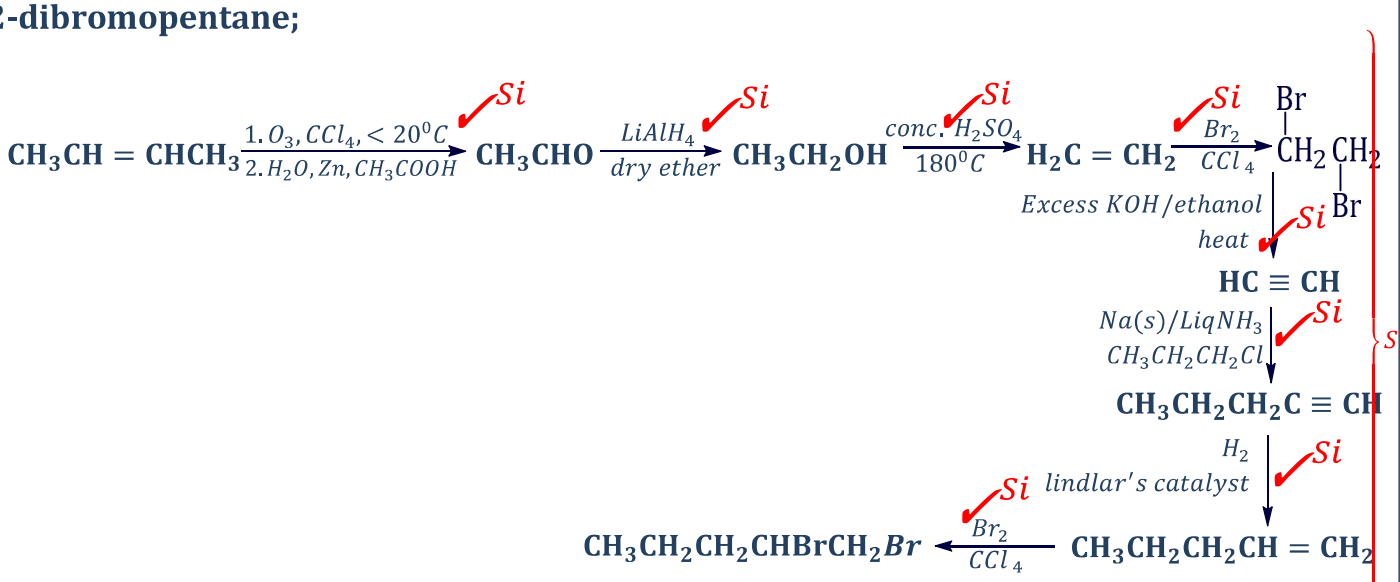
$2Mc = \text{Mc} = 02$ *Scores*

Synthetic **pathways to related products;**

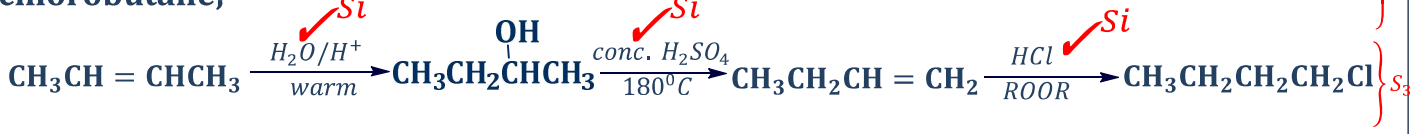
Benzoic acid,{  **};**



1,2-dibromopentane;



1-chlorobutane;



$S_1 + S_2 + S_3 = S = 11\ Scores$

Environmental impacts of producing above compounds in industry;

*When burnt, they give off poisonous gases such as carbon monoxide (CO), carbon dioxide (CO_2), hydrogen chloride (HCl), and hydrogen bromide, hence lead to **air pollution**, **acid rain**, **respiratory problems**, and **global warming**. The acid gases can also corrode metals and destroy plant leaves. Mitigated avoid **open-air burning** and instead use **safe chemical disposal methods**.*

di de dm

$1di + 1de + 1dm = D_3 = 03\ Scores$

6. **Formulae of P ;**

$\boldsymbol{Mass\ of\ carbondiode\ formed = 8.8g}$ $\boldsymbol{and\ mass\ of\ water\ formed = 1.8}$

$\boldsymbol{mass\ of\ carbon = \frac{12}{44} \times 8.8 = 2.4g}$

$\boldsymbol{mass\ hydrogen = \frac{2}{18} \times 1.8 = \ 0.2g}$

Element	C	H
%composition	*2.4*	*0.2*
Moles	$\frac{2.4}{12} = 0.2$	$\frac{0.2}{1} = 0.2$
Simplest ratio	$\frac{0.2}{0.2} = 1$	$\frac{0.2}{0.2} = 1$

CA

$\therefore$ *Empirical formula is* **CH**

Mass of p vaporised $= 0.100g$

At a Temperature $T = 273 + 273 = 546K$

At a Pressure $P = \left(\frac{734}{760} \times 1\right)\ atm$

At a Volume $V = 4.46 \times 10^{-2} dm^3$

Note: 1. *we use;* $\boldsymbol{R = 0.0821}$ L · atm/(mol · K), *when, P is in atm and V is in litre/*dm^3

2. *we use;* $\boldsymbol{R = 8.314\ J/(molK)}$, *when, P is in Pa and Volume is in* m^3

From ideal gas law $PV = nRT$

$$PV = \frac{m}{M_r}RT$$

$$\Longrightarrow M_r = \frac{mRT}{PV} = \frac{0.100 \times 0.0821 \times 546}{\left(\frac{734}{760} \times 1\right) \times 4.46 \times 10^{-2}} = \frac{4.48266}{0.04307} \approx \mathbf{104}g$$

CA

$\therefore$ $\boldsymbol{(CH)_n = 104}$

$(12 \times 1 \times n) + (1 \times 1 \times n) = 104$

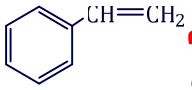
$13n = 104$

$n = \frac{104}{13} = 8$

$\therefore$ *Molecular formula is* $\mathbf{C_8H_8}$

CA

Structure and IUPAC Name;



CA

ethenylbenzene

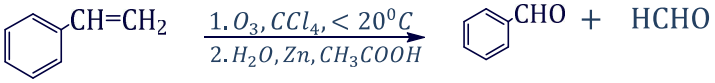
CA

Structural isomers;

P *has no other strural isomers.*

I

Identity of P and mixture Q;



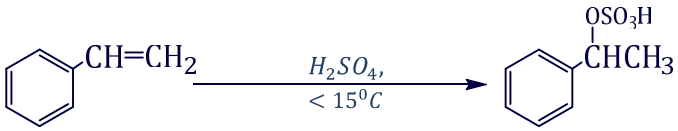
Q *is a mixture of benzylaldehyde and methanal with functional group* —C(=O)—H

I

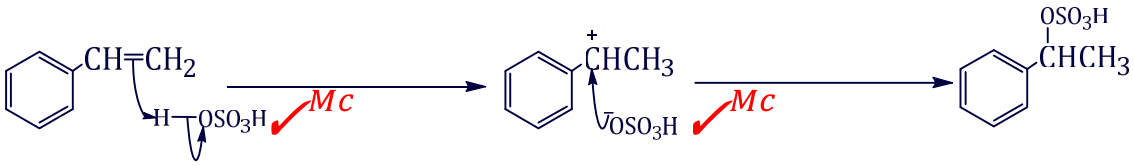
$5CA + 2I = F = 06\ Scores$

Mechanisms for reactions of P with bromine and concentrated sulphuric acid;

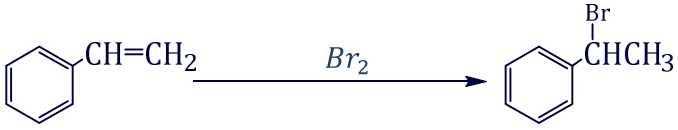
With concentrated sulphuric acid;

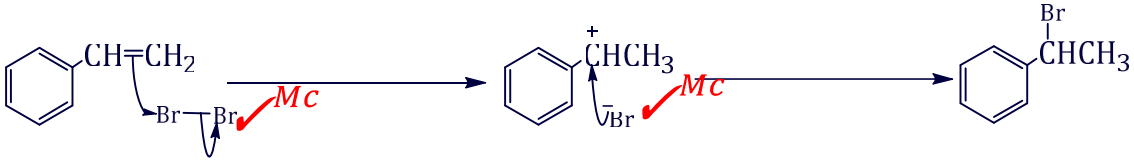


Mechanisim



With bromine;



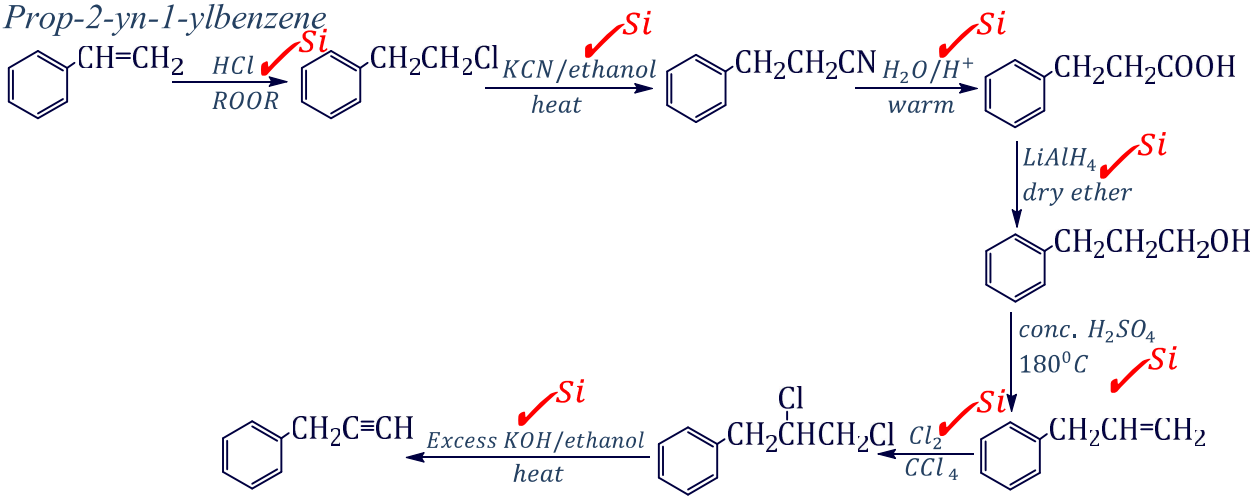


$M_C = 02$

$4Mc = Mc = 02\ Scores$

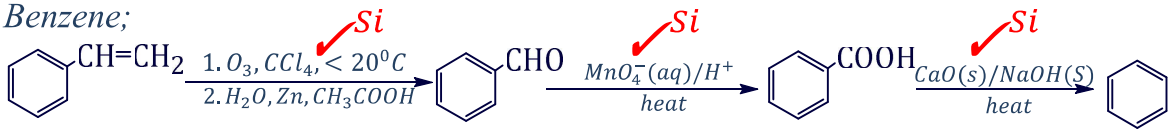
Synthetic pathways from Albertine oil fractions P to synthesis industrial compounds ;

Prop-2-yn-1-ylbenzene



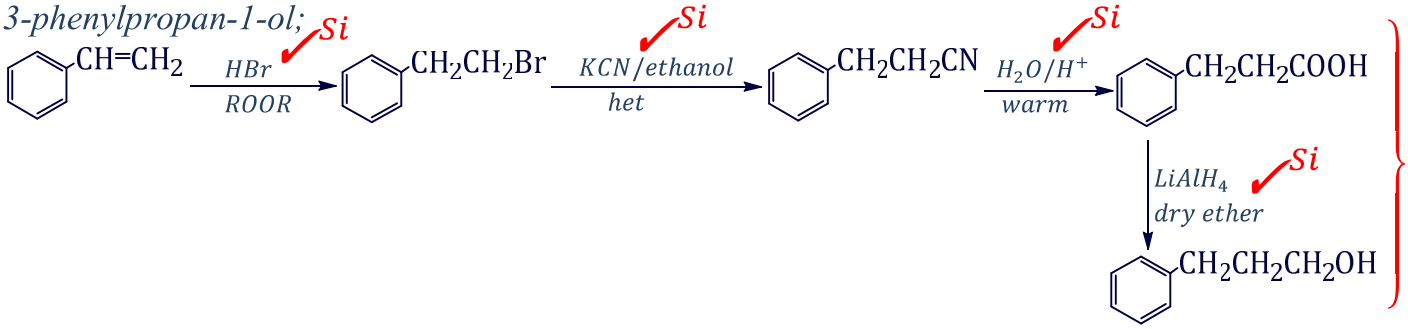
$S_1 = 05$

Benzene;



$S_3 = 01$

3-phenylpropan-1-ol;



Si Si Si Si

$S_1 = 05$

$S_1 + S_2 + S_3 = S = 03\ Scores$

Environmental impacts of producing above compounds in industry;

When burnt, they give off poisonous gases such as carbon monoxide (CO), carbon dioxide (CO_2), hydrogen chloride (HCl), and hydrogen bromide, hence lead to ***air pollution***, ***acid rain***, ***respiratory problems***, *and* ***global warming***. *The acid gases can also corrode metals and destroy plant leaves. Mitigated avoid* ***open-air burning*** *and instead use* ***safe chemical disposal methods***.

di de dm

1di+1de+1dm=D₃=03 Scores

END

2025

It is All about mindset change